VANISHING OF MAPS FROM A BOUNDED COMPLEX OF PROJECTIVES INTO A RIGHT-BOUNDED COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 7104-7118, following the statement at 7088-7102.

Editorial note. No omissions: both vanishing halves are identified with their reasons.

Allowed standard prerequisites (editorial).

- Maps from projective complexes. lemma-morphisms-from-projective-complex (same chapter, cited at ~7106). First equality Hom_K = Hom_D.
- Projectives into acyclic complexes. lemma-projective-into-acyclic-is-zero (same chapter, cited at ~7118). Second vanishing.

Source reference. Original lemma 064B: sources/stacks-derived/pinned-original-source.tex, source lines 7039–7039. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0647: sources/stacks-derived/pinned-original-source.tex, source lines 6966–6966. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original remark 08J5: sources/stacks-derived/pinned-original-source.tex, source lines 4293–4293. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{A}$ be an abelian category. Let $P^\bullet$, $K^\bullet$ be complexes. Let $n \in \mathbf{Z}$. Assume that*

(1) *$P^\bullet$ is a bounded complex consisting of projective objects,*
(2) *$P^i = 0$ for $i < n$, and*
(3) *$H^i(K^\bullet) = 0$ for $i \geq n$.*

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Then $\mathrm{Hom}_{K(\mathcal{A})}(P^\bullet, K^\bullet) = \mathrm{Hom}_{D(\mathcal{A})}(P^\bullet, K^\bullet) = 0$.

Proof. The first equality follows from Lemma 064B. Note that there is a distinguished triangle

$$(\tau_{\leq n-1}K^\bullet, K^\bullet, \tau_{\geq n}K^\bullet, f, g, h)$$

by Remark 08J5. Hence, by Lemma 0149 it suffices to prove $\mathrm{Hom}_{K(\mathcal{A})}(P^\bullet, \tau_{\leq n-1}K^\bullet) = 0$ and $\mathrm{Hom}_{K(\mathcal{A})}(P^\bullet, \tau_{\geq n}K^\bullet) = 0$. The first vanishing is trivial and the second is Lemma 0647. □